

## ✓ Milestone 5 | FastKitchen Customer Analysis

**INTRODUCTION:** FastKitchen is a fictitious restaurant and the dataset you'll be working with is constructed. While this dataset might not represent real data, it emulates characteristics of real data. When you're interviewing for a job, you might be asked to look at this kind of data to show off your skills in a context related to the company and the job position!

In this Milestone, you'll step into the shoes of a data contractor who is helping a new fast-food restaurant understand their customer base. You will need to make use of one type of outer join to help the restaurant manager combine information about their customers. These customers include registered customers who have accounts on the restaurant's website, and guest customers who do not register for accounts.

**HOW IT WORKS:** Follow the prompts in the questions below to investigate your data. Post your answers in the provided boxes: the **yellow boxes** for the queries you write, and **blue boxes** for text-based answers. When you're done, export your document as a pdf file and submit it on the Milestone page – see instructions for creating a PDF at the end of the Milestone. Please don't ever remove (paste your query below 📌) or (write your **answer** below 📌). These help your Evaluator!

### – Data Set **Description**

The data in this Milestone (`fastkitchen.*`) depicts orders made at a fictional takeout-only fast food restaurant in the Midwestern United States. The restaurant has an online site where customers can put in orders for carryout or delivery; customers can also make orders offline at the restaurant's storefront. You will be working with two tables in this Milestone: orders and users.

Each row in the `orders` table is a single order that was placed at the restaurant. This table has seven columns:

- `order_id` - unique order id, primary key

- **timestamp** - when the order was made
- **user\_id** - user\_id for registered accounts, blank if guest customer
- **order\_type** - whether the order was made onsite, online carryout, or online delivery
- **subtotal** - base amount for the order
- **tip** - amount of tip, if any, left by the customer
- **total** - subtotal + tip

Customers have the option of creating a user account, which can be used both in person and online. The `users` table has five columns:

- **user\_id** - unique user\_id value, primary key
  - **reg\_timestamp** - when the user registered their account
  - **city** - user city
  - **state** - two-letter code for state
  - **zip** - zip code
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## – Task 1: Explore information about orders.

To start off, let's warm up with some questions on the individual tables, before we ask questions that require joining the two tables together. Let's look at the `orders` table first.

**A.** What is the average total amount (including tips) spent per order?

(paste your query below 🖱️)

```
SELECT
  ROUND(AVG(total), 2) AS avgTotalAmount
FROM
  fastkitchen.orders
```

(write your **answer** below 🖱️)

\$22.22

- B. Compare the average subtotals, tips, and totals spent by each order type (onsite, carryout, delivery).

(paste your query below 📌)

```
SELECT
  order_type,
  ROUND(AVG(subtotal), 2) AS avgSubtotal,
  ROUND(AVG(tip), 2) AS avgTip,
  ROUND(AVG(total), 2) AS avgTotalAmount
FROM
  fastkitchen.orders
GROUP BY
  order_type
```

Are there any major differences between order types?

(write your **answer** below 📌)

There are no major differences between order types, average total amount is highest for delivery because of the average subtotal being little over 40 cents more than the rest but that is about it and carryout has highest tip with a 7 to 14 cent margin.

- C. Write a query to count the number of orders made by **registered users**.

(paste your query below 📌)

```
SELECT
  COUNT(user_id) AS registered_users
```

```
FROM
  fastkitchen.orders
WHERE
  user_id IS NOT NULL
```

- D. Now, write a query to return the number of orders made by **non-registered customers**. Remember, non-registered customers don't have a user id.

(paste your query below 📌)

```
SELECT
  COUNT(*) AS nonregistered_users
FROM
  fastkitchen.orders
WHERE
  user_id is NULL
```

Compared alongside the returned count from the query in C, which group is larger?

(write your **answer** below 📌)

Guest users are larger with a count of 2088 and registered 1932.

- E. While guest customers tend to place more orders, there might be underlying reasons behind this trend. Your goal is to understand user behavior and suggest a practical solution to help improve customer registration and retention.



**Try this prompt:** Guest customers placed more orders than registered users. Why might guests order more often, and what could FastKitchen do to encourage guests to register accounts without reducing conversion?

Based on ChatGPT's response, what's one potential reason for high guest activity? What's one idea FastKitchen could try to convert guest users into registered users?

(write your **answer** below 🖊)

The main reason for high guest activity could be ordering as a guest is more convenient since it takes less effort and steps to order without an account and have faster checkout. One way to encourage guest users to register is to offer incentives such as discounts on future orders or faster checkouts for registered users.

## – Task 2: Explore information about registered users.

Next, we'll check out the `users` table.

**A.** Write a query that counts the number of users by city.

(paste your query below 🖊)

```
SELECT
  city,
  COUNT(*) AS n_users
FROM
  fastkitchen.users
GROUP BY
  city
```

Which city has the highest number of users, and how many users are there?

(write your **answer** below 📎)

Allen, 212

- B.** Expand the query so that you group by zip code as well. Does this help explain what you found in part 2A?

(paste your query below 📎)

```
SELECT
  city,
  zip,
  COUNT(*) AS n_users
FROM
  fastkitchen.users
GROUP BY
  city, zip
```

(write your **answer** below 📎)

Yes this query shows Allen being distributed largely over multiple zip codes while the other cities only have one zip code.

### – **Task 3:** How do orders compare between zip codes and cities?

Finally, we'll combine the `user` and `orders` tables into a single, joined table.

- A.** To start, simply write a query that returns all of the columns, joining the two tables on the `user_id` column. Make sure that you choose a join that keeps all of the orders, even when there isn't a matching registered user.



**Try this prompt:** I'm trying to join an orders table with a users table on user\_id in my SQL query, but not all orders have a matching user (some are guests). Which type of JOIN should I use if I want to keep all orders, even when there's no matching user, and why?

(paste your query below 📌)

```
SELECT
  *
FROM
  fastkitchen.orders AS a
LEFT JOIN fastkitchen.users AS b
ON a.user_id = b.user_id
```

- B.** Add to the query from 3A to answer the following question: in which zip code is the user with the highest amount of money spent?

(paste your query below 📌)

```
SELECT
  b.zip,
  SUM(a.total) AS total_spent
FROM
  fastkitchen.orders AS a
LEFT JOIN fastkitchen.users AS b ON a.user_id = b.user_id
WHERE
  a.user_id IS NOT NULL
GROUP BY
  b.zip,
  a.user_id
ORDER BY
  total_spent DESC
LIMIT
  1
```

(write your **answer** below 📌)

63222, with total of \$1103.11

- C. Write a query that returns the average total amount spent per order by zip code.

(paste your query below 📌)

```
SELECT
  b.zip,
  ROUND(AVG(a.total), 2) AS average_total
FROM
  fastkitchen.orders AS a
  LEFT JOIN fastkitchen.users AS b ON a.user_id = b.user_id
WHERE
  a.user_id IS NOT NULL
GROUP BY
  b.zip
```

How many of the zip codes spend more on average than non-registered guest customers?

(write your **answer** below 📌)

Three zip codes spend more on average than non registered guest customers.

## – Submission

Great work completing this Milestone! To submit your completed Milestone, you will need to download / export this document as a PDF and then upload it to the



Milestone submission page. You can find the option to download as a PDF from the File menu in the upper-left corner of the Google Doc interface.